Tactix: Sweep Gate and Learning-Readiness Checkpoint

Report to Andrew

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Outcome

The AM427 Sweep scheduling candidate remains unaccepted. Matched Tier A and B are independently certificate-clean, but neither establishes N4 parity or a stable performance gain. Tier C is still running on the same pinned binaries; no Tier D comparison has been submitted. A source review confirms that production online learning needs policy/session refactoring, while the existing root-action executor supports bounded opt-in prototypes. The AM429 and AM430 technical reports hold the detailed evidence.

Verified evidence

Tier A: both arms solved 199/200, including 117 independently checked native DSR proofs each, with identical paired verdicts and proof sizes. AM427 took 3.0 more solver seconds across 200 tasks. Tier B: both arms solved 28/28 (14 SAT, 14 UNSAT), with all 14 proofs and all models checked; paired verdicts and proof sizes again matched. AM427 took 415.978 versus the parent's 427.866 solver seconds over 28 tasks, a small, non-gating advantage. The first Tier C submission never started because the active B build lock refused acquisition; the general runner now holds an array until its locks and dependent aggregate are established. Its 31 focused runner/oracle tests pass.

As of 18:41 UTC, corrected Tier C array `1723299` had 46/60 task records (23 complete pairs): 30 SAT and 16 UNSAT, no recorded model/proof failure or expected-verdict mismatch, and identical paired verdicts and proof sizes. The remaining tasks and aggregate `1723300` are active; these partial records are not an aggregate performance result. The checker bundle is frozen by SHA-256 in the run manifest. No mutable checker clone is used.

The fixed ASan+UBSan public-action/online suite passed 10/10; a separately rebuilt six-test API/online subset and the scheduling test passed. The latter had a parent-identical stale witness: early Lucky refuted its old tiny binary formula before the collection boundary under test. A deeper independent UNSAT tail restores the test's intended path. `DefaultPolicy` and its scheduler are now non-copyable/non-movable because their arms borrow configuration or member addresses; this changes no solver schedule.

Learning design and open gates

For inprocessing order/dose, the explicit online executor already has typed steps, limits and replay, but the mature default scheduler separately owns clocks, admission, cursors and follow-ups. A shared default-exact slot controller is the first major refactor. For branching and clause deletion, current sub-policy pointers are borrowed and cover only part of the decision; owned typed arms, safe switch boundaries, state migration and complete reduction-strategy control are the second. A third necessary seam is session lifetime: `tactix_solve` is synchronous and constructs a fresh `DefaultPolicy` per call, so no live default-policy arm can yet be reconfigured through the C API. Preserve default action/proof traces and measure overhead before enabling a learner.

N2 is still open: the frozen canonical no-option DSR checker rejects the separate H2 proof; the implied-unit deletion option is only a diagnostic alternative with a different proof-state contract. Broad affected fuzz, N1 completeness, N3/Tier-D competitiveness, N4 reference scheduling and the retained Rank/B+C/WV gates remain open. The proof-rule audit found a qualified winner gap in general live pseudo-Boolean arithmetic/redundancy/dominance; ordinary CaDiCaL/Kissat clausal emitter events map to existing RUP/RAT/delete. Finish Tier C and affected fuzz, settle and fully audit the H2 contract, then run the cumulative gates. Only after those fixes should the 1,166-instance Tier D pass use the frozen winner allocation: four CPUs, 32 GiB and 5,000 solver seconds per task on `cpu-dense-hwulab-q`.